IMPERIAL COLLEGE LONDON

DEPARTMENT OF AERONAUTICS

Thermodynamics Tutorial Sheet 1

Treat **air** as a perfect gas throughout, with R = 287 J/(kg K), γ = 1.4

Kinetic Theory Review

Question 1

The Knudsen number for flow at high altitude about a body is 0.1 for conditions where ambient pressure is 1 Pa and temperature is 200 K. How would the Knudsen number change in a region where pressure is 100 Pa and temperature is 270 K?

Answer: times a factor of 0.0074

$$Kn = \frac{\lambda}{L} = 0.1 \text{ at high altitude}$$

$\therefore$ The flow is not continuum

$$\because Kn = \frac{\lambda}{L} = \frac{1}{\sqrt{2} n \sigma L} = \frac{V}{\sqrt{2} N \sigma L}$$

$$\because PV = MRT \qquad \therefore V = \frac{MRT}{P}$$

$$\therefore Kn = \frac{MRT}{\sqrt{2} \sigma N L P} \propto \frac{T}{P}$$

$$\therefore \frac{Kn}{Kn'} = \frac{\left(\frac{T}{P}\right)}{\left(\frac{T'}{P'}\right)} = \frac{TP'}{T'P} = 74.074$$

$$\therefore Kn' = \frac{Kn}{74.074} = 1.35 \times 10^{-3}$$

Question 2

(a) Calculate the mass of hydrogen gas required to yield 500 GJ by combustion in air. Assume a calorific value of 120 MJ/kg and a gas constant of 4.12 kJ kg^{-1}K^{-1} for hydrogen. (This is the lower heating value; the higher (HHV) value is only achieved if the energy lost in the water vapour product is recovered by condensation).

(b) Calculate the volume occupied by the mass of hydrogen in part (a) assuming it is compressed to 350 bar at 15ºC.

(c) Calculate the ratio of the volume occupied by hydrogen in part (b) to that required by the quantity of Jet A-1 aviation fuel that would yield the same energy by combustion in air. Assume a calorific value of 43 MJ/kg and a density of 800 kg.m-3 for Jet A-1.

(d) Determine the work required to compress 1 kg of hydrogen from 1 bar to 350 bar, assuming the temperature is maintained constant at $15^{\circ}\mathrm{C}$.

Question 1 is mostly about making sure you use the correct units.

(a) $$M = \frac{500\ \mathrm{GJ}}{120\ \mathrm{MJ/kg}} = 4.1\dot{6} \times 10^3\ \mathrm{kg}$$

(b) $$V = \frac{MRT}{P} = 141.25\ \mathrm{m^3}$$

(c) $$V_f = \frac{500\ \mathrm{GJ}}{43\ \mathrm{MJ/kg} \times 800\ \mathrm{kg/m^3}} = 14.535\ \mathrm{m^3}$$

$$\therefore \frac{V_f}{V} \times 100\% = 10.29\%$$

(d) $$W = \int P\,dV = \int \frac{MRT}{V}\,dV = MRT \int \frac{1}{V}\,dV$$

$$= MRT \ln\left|\frac{V_2}{V_1}\right| = MRT \ln\left|\frac{P_1}{P_2}\right| \quad \text{as } P \propto \frac{1}{V}$$

$$= -6.95 \times 10^6\ \mathrm{J}$$

Question 3

A radioisotope power source (RPS) is designed to utilize the Americium isotope ^{241}Am in the form of Americium dioxide AmO_2 as a heat source for a deep space satellite. The RPS is to be used in conjunction with a Stirling cycle engine for electrical power generation. Assume the overall efficiency of power conversion to be 30% and the properties below.

Estimate the mass of americium required and the minimum volume (excluding shielding) required to provide a continuous 50 W electrical power. By how much will the power output have decreased after 10 years?

^{241}Am decays with majority of energy associated with 5.5MeV α-particles

Assume the source has specific activity 1.27×10^{11} Bq/g,

density 11.7 g/cm^3, half-life 432 years.

Note: you do not need to know anything about the Stirling cycle; if you are interested search up "Advanced Stirling Radioisotope generator".

$$P = \frac{P_t}{30\%} = \frac{500}{3}\,W$$

$$\therefore M_{min} = \frac{P}{AE} = 1.49\,kg$$

$$\therefore V_{min} = \frac{M_{min}}{\rho} = 1.27 \times 10^{-4}\,m^3$$

After 10 years:

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} \qquad \because P = MAE \quad \therefore P \propto A$$

$$\therefore P' = Pe^{-\lambda t} = Pe^{-\frac{\ln 2}{t_{\frac{1}{2}}}t} = 164.01\,W$$

$$\therefore \Delta P = P - P' = 2.65\,W$$

Question 4.

A conventional hot air balloon contains air at 100 ºC, whilst the surrounding atmosphere is at 1 bar and 20 ºC. It can be assumed that the balloon is fully inflated, so the volume contained in the balloon envelope is constant.

(i) For the conditions given, calculate the volume of air inside the envelope required per kg of lift.

(ii) Difficult: Explain why increasing the temperature of the balloon by lighting the burner can be modeled as a constant pressure heat addition.

(i) $L = mg = \rho V g = (\rho_{air} - \rho_{ballon}) V g$

$\therefore M = (\rho_{air} - \rho_{ballon}) V \qquad \because P = \rho R T$

$$\therefore \frac{V}{M} = \frac{1}{\rho_{air} - \rho_{ballon}}$$

$$= \frac{1}{\frac{P}{RT_{air}} - \frac{P}{RT_{ballon}}} = 3.92\ m^3/kg$$

(ii) At constant pressure and no work is done since air could flow out the ballon:

$$dQ = mC_p\, dT + 0 = \frac{PV}{RT} C_p\, dT = \frac{PVC_p}{R}\left(\frac{1}{T} dT\right)$$

As T change from T to T':

$$\Delta Q = \frac{PVC_p}{R} \ln\left|\frac{T'}{T}\right| = \frac{PVC_p}{R} \ln\left|\frac{T+\Delta T}{T}\right|$$

$$= \frac{PVC_p}{R} \ln\left|1 + \frac{\Delta T}{T}\right|$$

Question 4.

A conventional hot air balloon contains air at 100 ºC, whilst the surrounding atmosphere is at 1 bar and 20 ºC. It can be assumed that the balloon is fully inflated, so the volume contained in the balloon envelope is constant.

(i) For the conditions given, calculate the volume of air inside the envelope required per kg of lift.

(ii) Difficult: Explain why increasing the temperature of the balloon by lighting the burner can be modeled as a constant pressure heat addition.

By common Maclaurin expansions:

$$\Delta Q = \frac{PVC_p}{R}\left(\frac{\Delta T}{T} - \frac{1}{2}\left(\frac{\Delta T}{T}\right)^2 + \frac{1}{3}\left(\frac{\Delta T}{T}\right)^3 \ldots\right)$$

Since ΔT is relatively small,

$$\Delta Q \approx \frac{PVC_p}{R}\left(\frac{\Delta T}{T}\right)$$

$$\approx \frac{PV}{RT} C_p \Delta T$$

$$\approx MC_p \Delta T$$

Question 5.

Work transfer W has been carefully defined in lectures. According to that definition, decide whether the *system* underlined in the examples below has done positive, negative or zero work in the processes indicated:

a) **Air,** initially at high pressure in a rigid bottle, is released slowly into the atmosphere until the pressure falls to that of the atmosphere.

b) An evacuated rigid bottle is filled by **air** which enters it when a tap is opened.

c) An evacuated rigid bottle is filled from a rigid bottle containing **air** at high pressure.

d) An **air/fuel mixture** passes through a stationary jet engine mounted on a test stand.

(a) +

(b) −

(c) 0

(d) ~~0~~ +

compressed → expanded → large expanded

Question 6.

A balloon of light flexible material is to be filled with air from rigid storage bottle until it has a volume of $0.5m^3$. If the balloon material requires negligible stress to stretch it, and the atmospheric pressure is $0.1013\ MN/m^2$, how much work is done by the air initially in the bottle? Ans: 50.6 kJ

$$W = \Delta PV = 0.1013\,MN/m^2 \cdot 0.5m^3$$

$$= 50650J$$

Question 7

A system executes a cyclic process during which there are four heat transfers, namely Q1 = 224kJ, Q2 = 20kJ, Q3 = -200kJ, Q4 = 37 kJ. There are four corresponding work transfers W1 = 50 kJ, W2 = -40kJ, W3 = 75kJ, and W4 = ?

$$\Delta E = 0$$

$$= Q_1 + Q_2 + Q_3 + Q_4 - W_1 - W_2 - W_3 - W_4$$

$$= 81\,kJ - 85\,kJ - W_4$$

$$\therefore W_4 = -4\,kJ$$

Question 8

A system consisting of a working fluid enclosed in a cylinder receives heat through the walls and does work by pushing out a piston.

At the start of the process the temperature is 100°C, and at the end, 200°C. The heat transfer during the process is given by

$$\frac{dQ}{dT} = 0.5\, kJ/K$$

and the work transfer by

$$\frac{dW}{dT} = [1-(0.001)T]\, kJ/K$$

What is the increase of the energy of the system during the process? Ans: -7.7 kJ

$$\Delta E = Q - W$$

$$= \frac{dQ}{dT}\Delta T - W$$

$$= 0.5\, kJ/k \times 100\, k - [T - 0.0005T^2]_{100+273}^{200+273}$$

$$= 50\, kJ - (361.1355\, kJ - 303.4355\, kJ)$$

$$= -7.7\, kJ$$

Question 9

One kg. of air is expanded slowly in a certain process from 7 bar and 38°C to 3.5 bar and 38°C. It does 15.3 kJ of work against a **frictionless** piston.

a) What is the increase of energy of the air in the expansion?

b) What is the heat transfer?

c) Is the process at all times isothermal? (Find out what W would be for an isothermal process and compare.)

Ans: part b) 15.3 kJ

a) $\Delta E = 0J$

b) $\Delta E = Q - W$

$\therefore Q = W = 15.3kJ$

c) No, cause ~~pressure changes~~ T might change in processes and friction exist in process although it expand slowly

For isothermal process:

$$W = \int P\,dV = \int \frac{MRT}{V}\,dV$$

$$= MRT \int \frac{1}{V}\,dV$$

$$= MRT \ln\left|\frac{V'}{V}\right| \qquad \because V = \frac{MRT}{P} \qquad \therefore V \propto \frac{1}{P}$$

$$\therefore W = MRT \ln\left|\frac{P}{P'}\right| = 61868.2379J$$

Question 10.

The specific heat at constant pressure of a certain substance is a function of a temperature only, given by

$$C_P = 2.09 + 42/(T - 200) \quad kJ.kg^{-1}.K^{-1}$$

One kg of the substance is heated as its volume increases from $2.0m^3$ to $2.4m^3$, the pressure being one bar and the temperature rising from 300°C to 450°C.

a) What is the heat transfer to the substance?
b) What is the change of energy?

Ans: a) 327.7kJ

b) 287 kJ

a) $\overset{Q}{\Delta U} = mC_p \Delta T$

$$= m\int_{573}^{723} \left(2.09 + \frac{42}{T-200}\right) dT$$

$$= \left[2.09T + 42\ln|T-200|\right]_{573}^{723}$$

$$= 327.7\,kJ$$

b) $\Delta E = \overset{Q}{\Delta U} - W$

$$= \Delta U - P\Delta V$$

$$= 287.7\,kJ$$